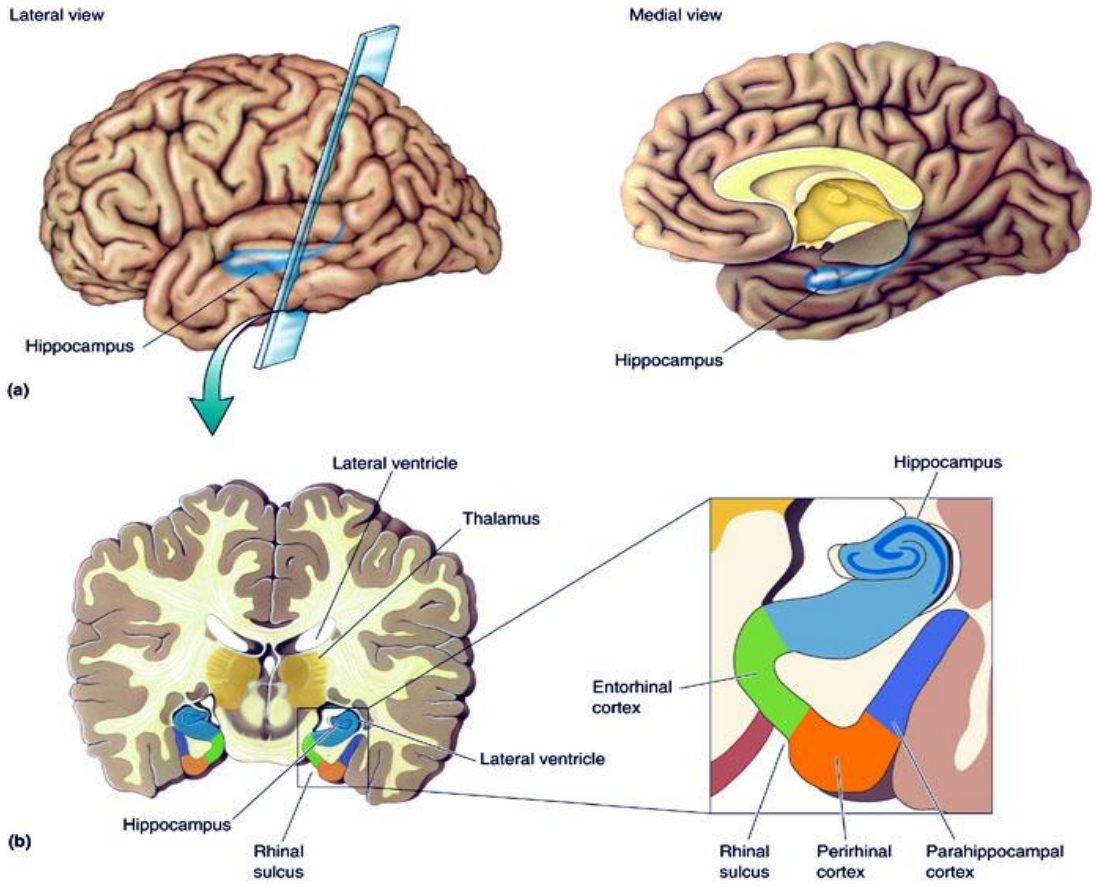
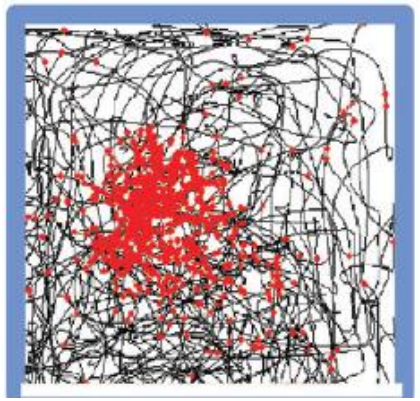
Brain-Like Replay Naturally Emerges in Reinforcement Learning Agents

Hippocampus as a memory and navigation machine

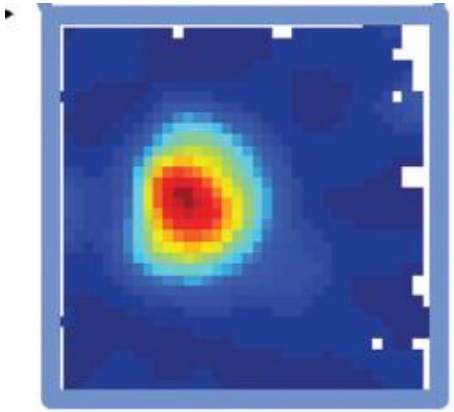


Bear, M., Connors, B., & Paradiso, M. A. (2020). *Neuroscience: exploring the brain, enhanced edition: exploring the brain*. Jones & Bartlett Learning.

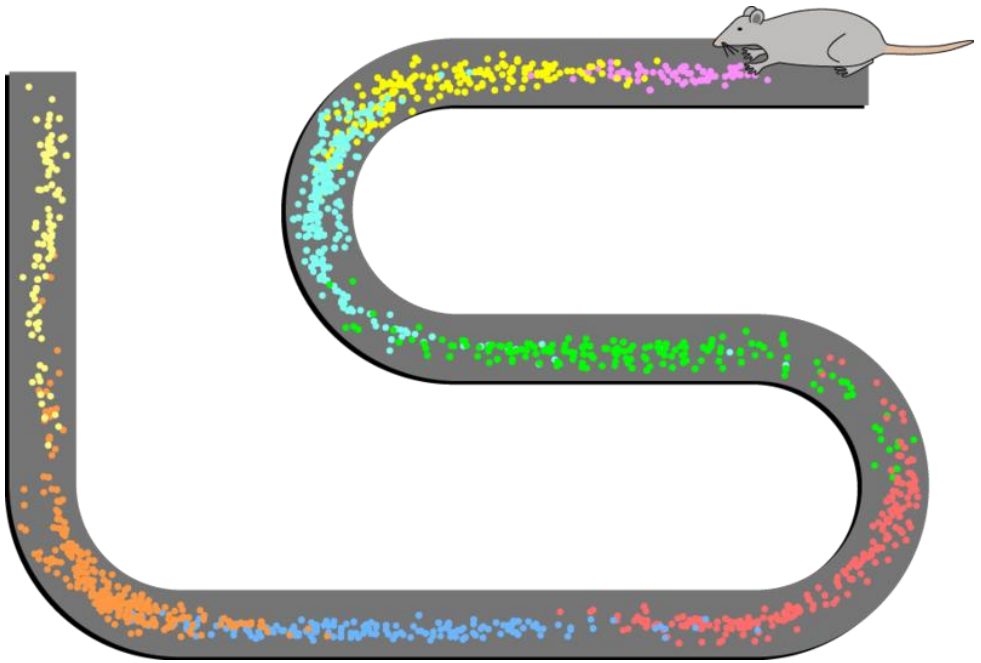
Roles in Navigation



Trajectory&
Place cell's
firing rate



Place field

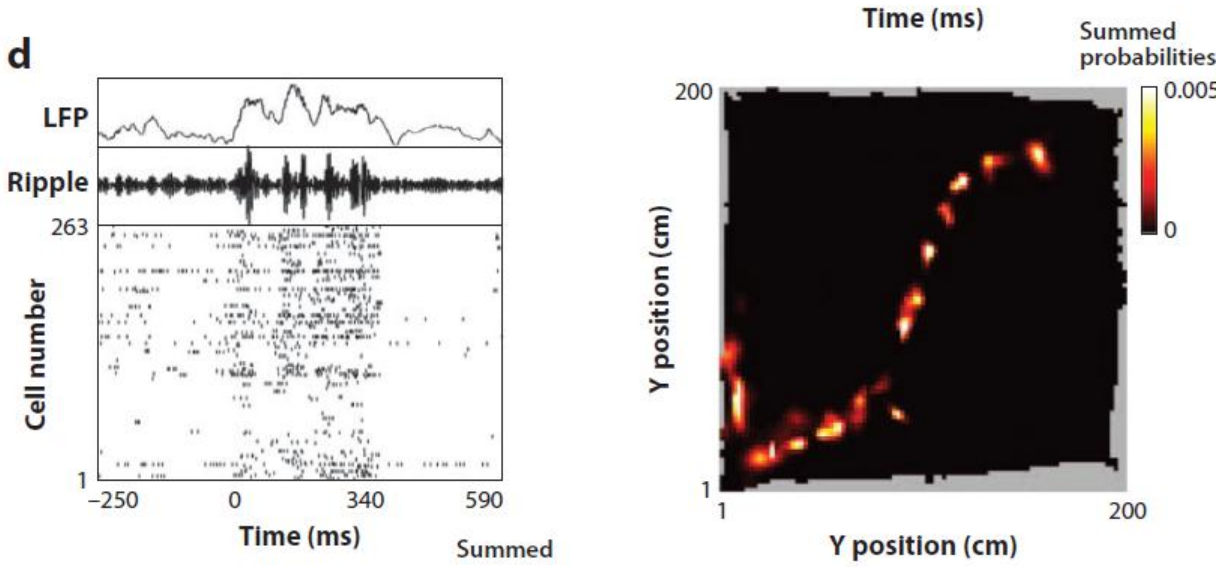


The place cells constitute **a cognitive map** through which navigation is possible and efficient.

Moser, M. B., Rowland, D. C., & Moser, E. I. (2015). Place cells, grid cells, and memory. *Cold Spring Harbor perspectives in biology*, *7*(2), a021808.

By Stuartlayton at English Wikipedia, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=43746578

Replay



Definition:

Reactivation of a sequence pattern of place cells that once fired during movement

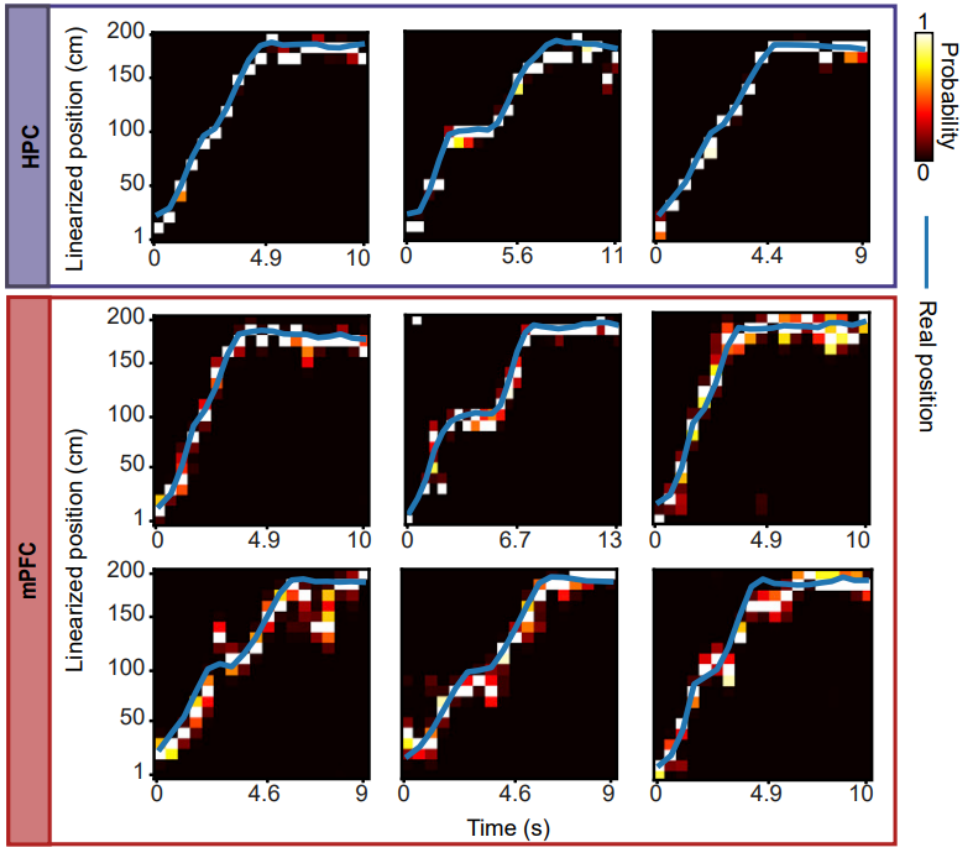
Foster, D. J. (2017). Replay comes of age. *Annual review of neuroscience*, *40*, 581-602.

"Reexperiencing"/"Organizing"

Putative functions

- **Memory consolidation/ Memory retrieval/ Spatial memory** (overlap with previous taken path)
- **Planning/ Navigation** (overlap with future taken path)

Replay also exists in cortex



Cortex replays are often coordinated with hippocampal replays.

Further support memory consolidation and navigation…

Hippocampo–prefrontal coordination / phase locking

Peyrache, A., Khamassi, M., Benchenane, K., Wiener, S. I., & Battaglia, F. P. (2009). Replay of rule-learning related neural patterns in the prefrontal cortex during sleep. *Nature neuroscience*, *12*(7), 919-926.

Current Research

(1)Theories relies heavily on their hypotheses , which have been complex and less convincing.

(2)Experimental results are solid themselves, but they often support different functions and contradict each other.

Our Goal and Method

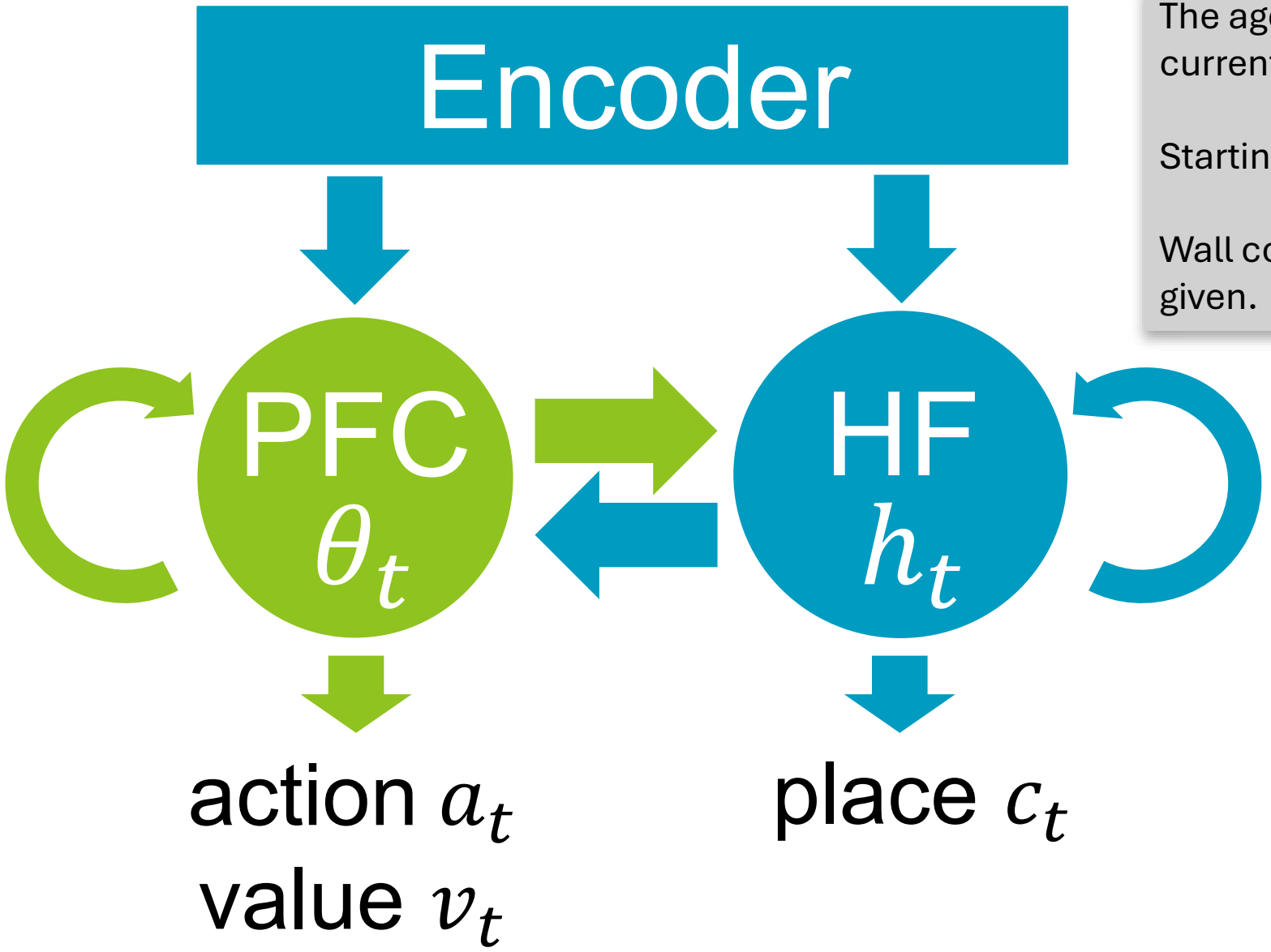
Let biological representations emerge as a natural result of task-optimized (goal-directed) training.

Deep reinforcement learning (DRL) is employed with the minimal sufficient conditions:

(1) Replay serves for reward maximization.

(2) Replay is a form of communication between the neocortex and hippocampus.

$$o_t = mask(s_t), a_{t-1}, r_{t-1}$$



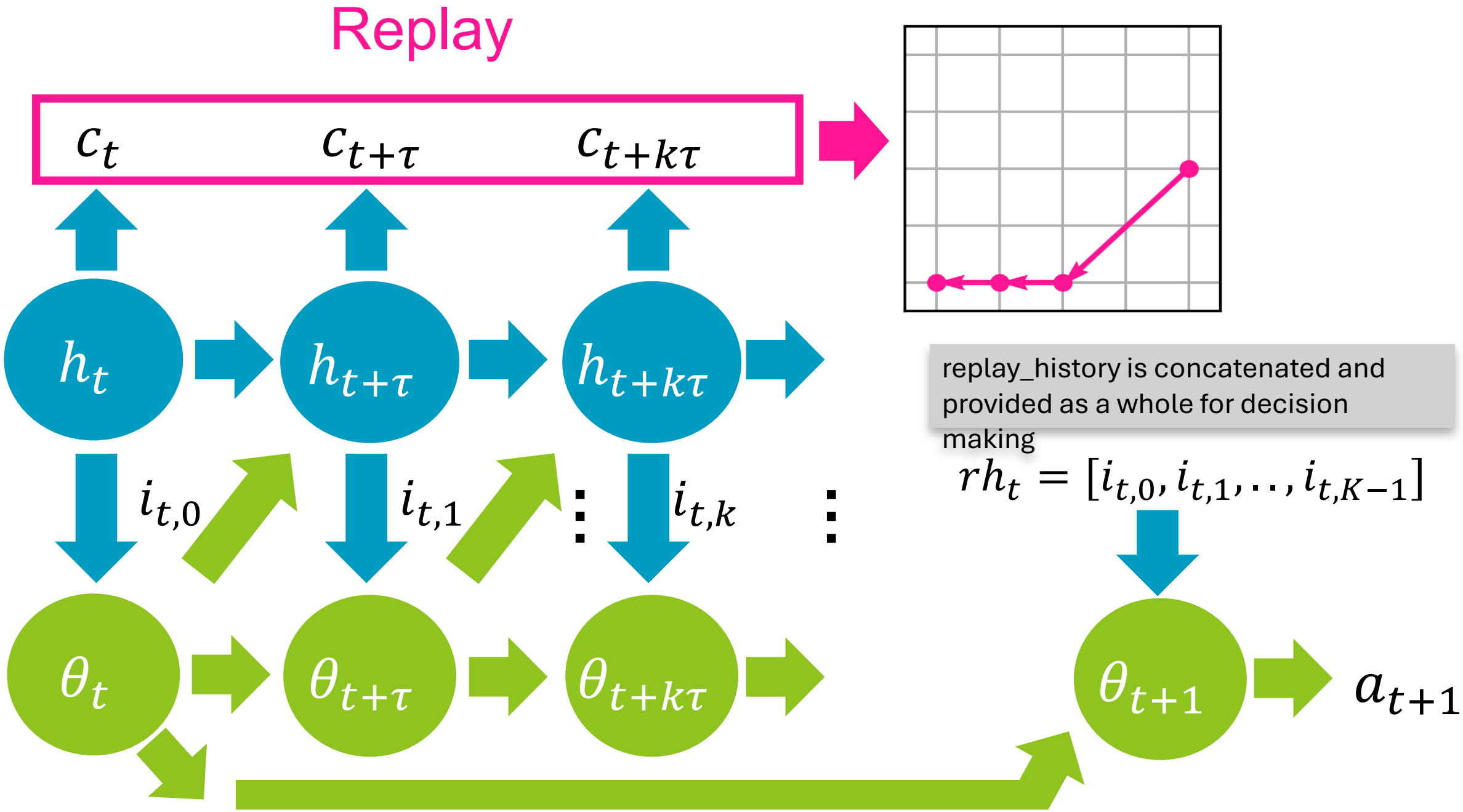
The agent only receives its current ***s*** in 10% time.

Starting ***s*** is always given.

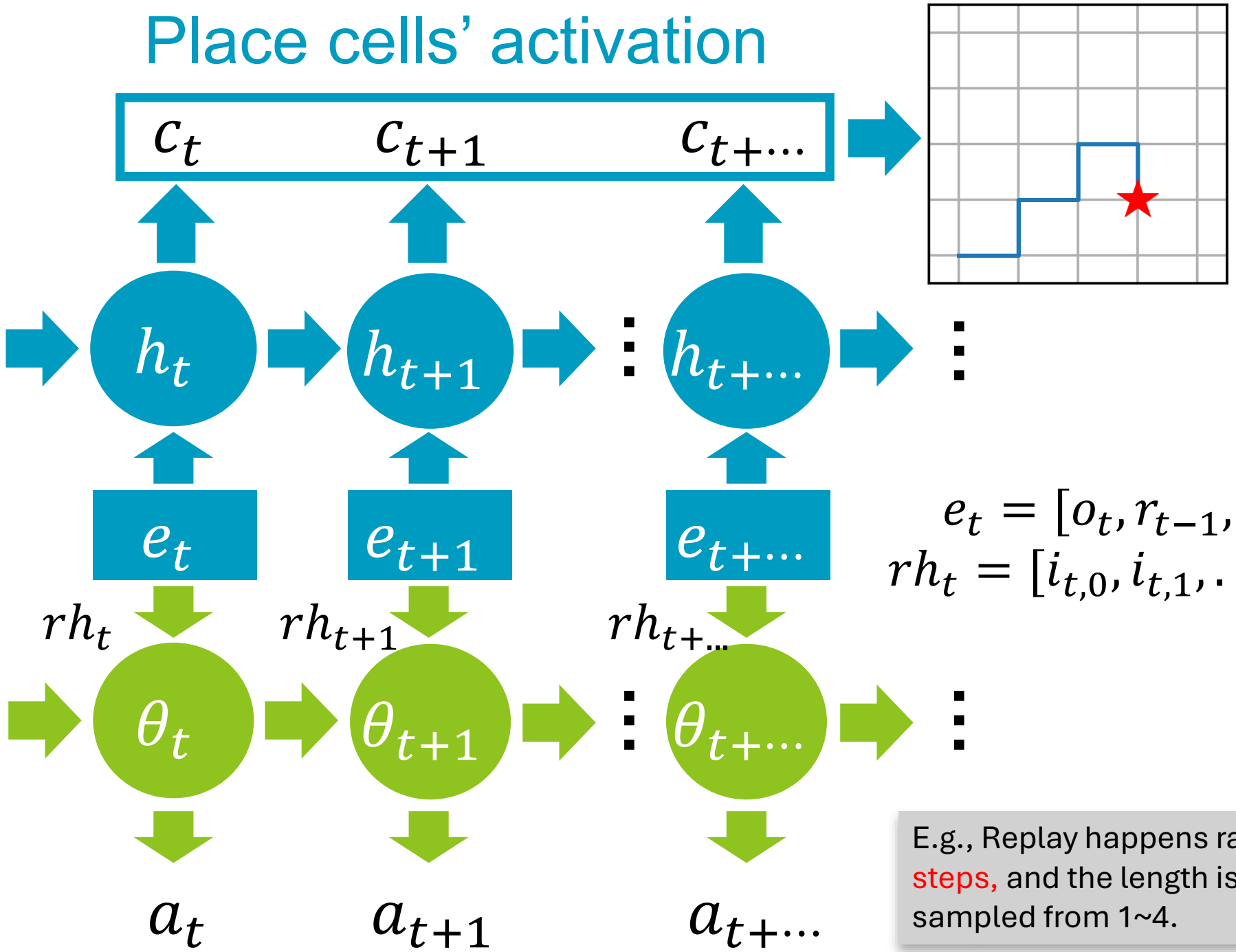
Wall configuration is always given.

Goal:
Maximizing the reward

Goal:
Predicting next locations


Replay
c_t
$c_{t+\tau}$
$c_{t+k\tau}$
h_t
$h_{t+\tau}$
$h_{t+k\tau}$
$i_{t,0}$
$i_{t,1}$
$i_{t,k}$
θ_t
$\theta_{t+\tau}$
$\theta_{t+k\tau}$
θ_{t+1}
a_{t+1}
replay_history is concatenated and provided as a whole for decision making
$rh_t = [i_{t,0}, i_{t,1}, .., i_{t,K-1}]$

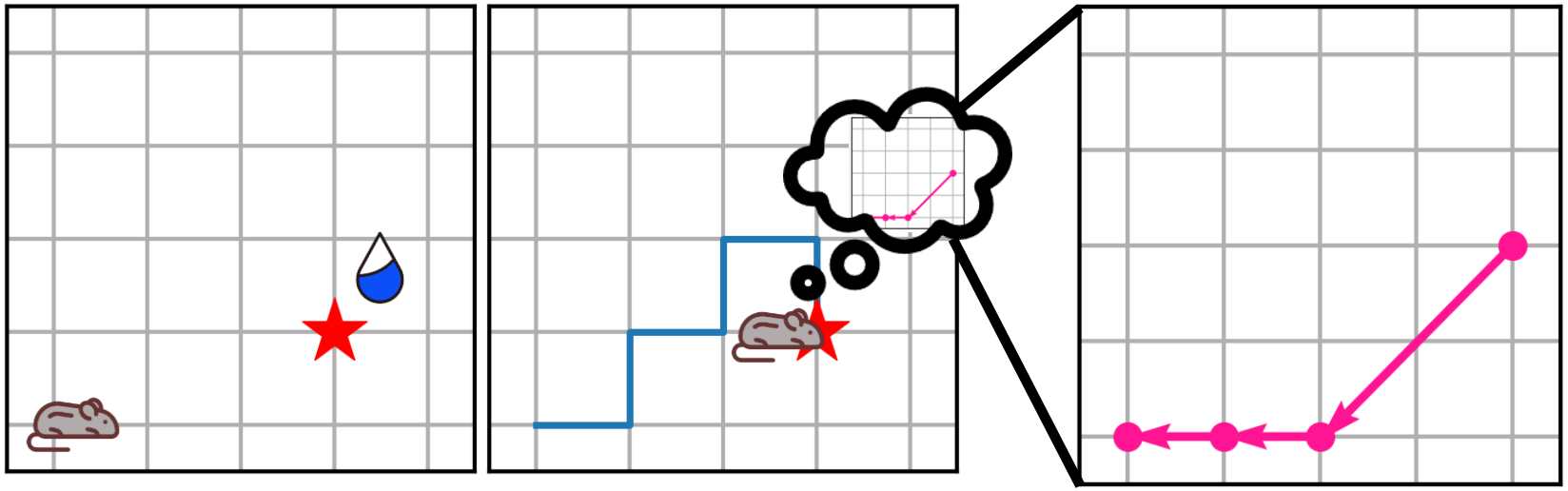
Place cells' activation

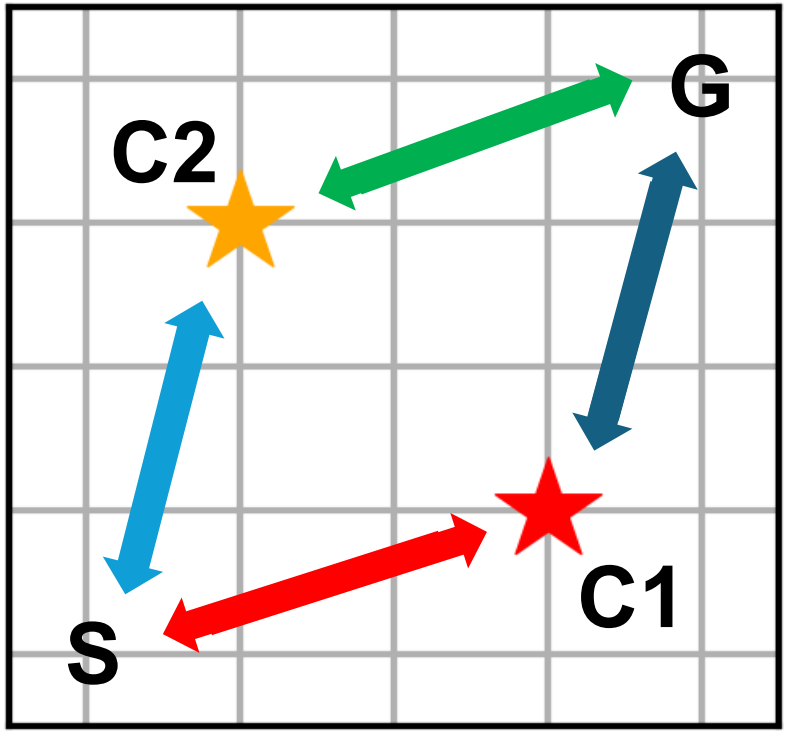


$$e_t = [o_t, r_{t-1}, a_{t-1}]$$
$$rh_t = [i_{t,0}, i_{t,1}, .., i_{t,K-1}]$$

E.g., Replay happens randomly at 50% steps, and the length is randomly sampled from 1~4.

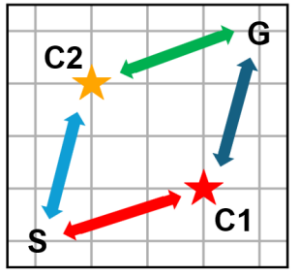
Experiment 1

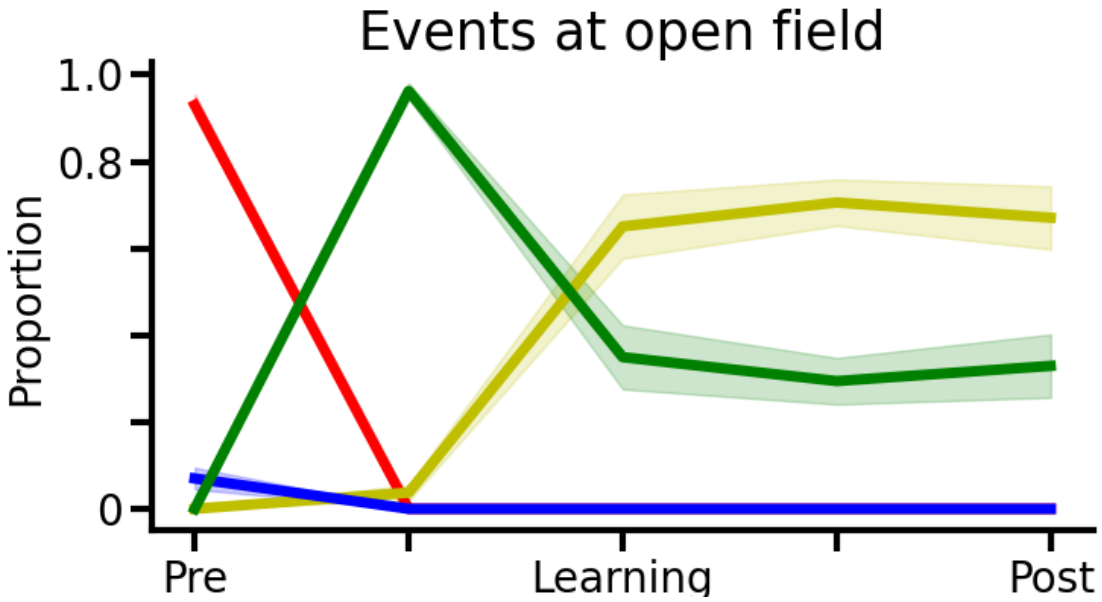


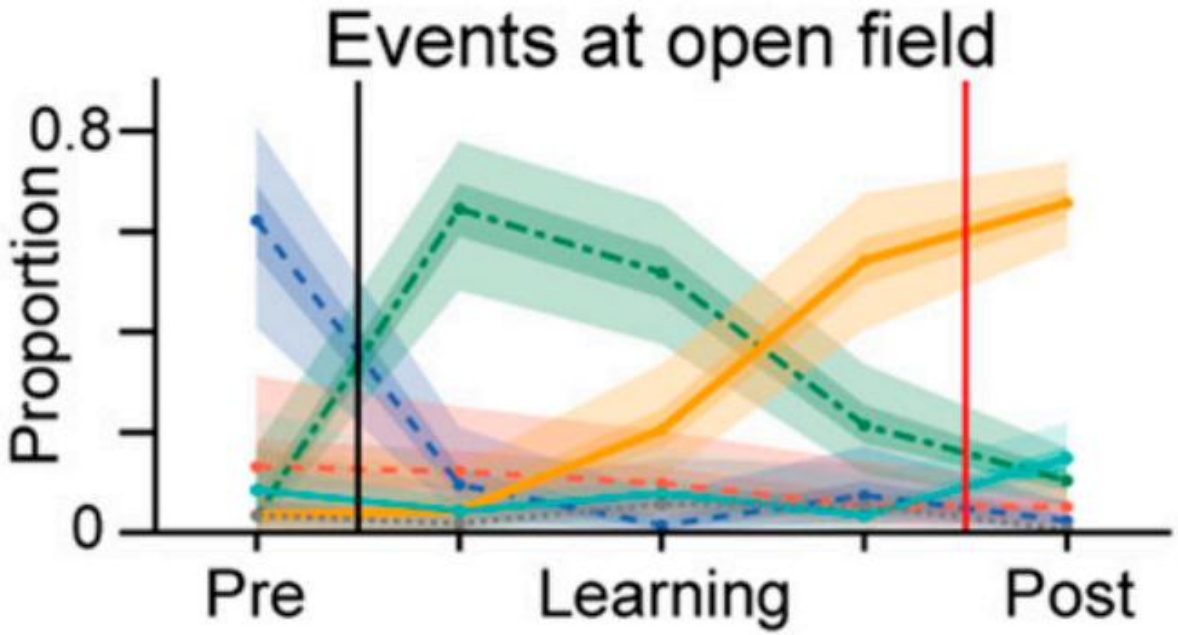


The blue line represents the replay trajectory;
The red line represents last replay trajectory when it reaches the goal.
The number of replay steps are set to be 4 after 1 step of action.

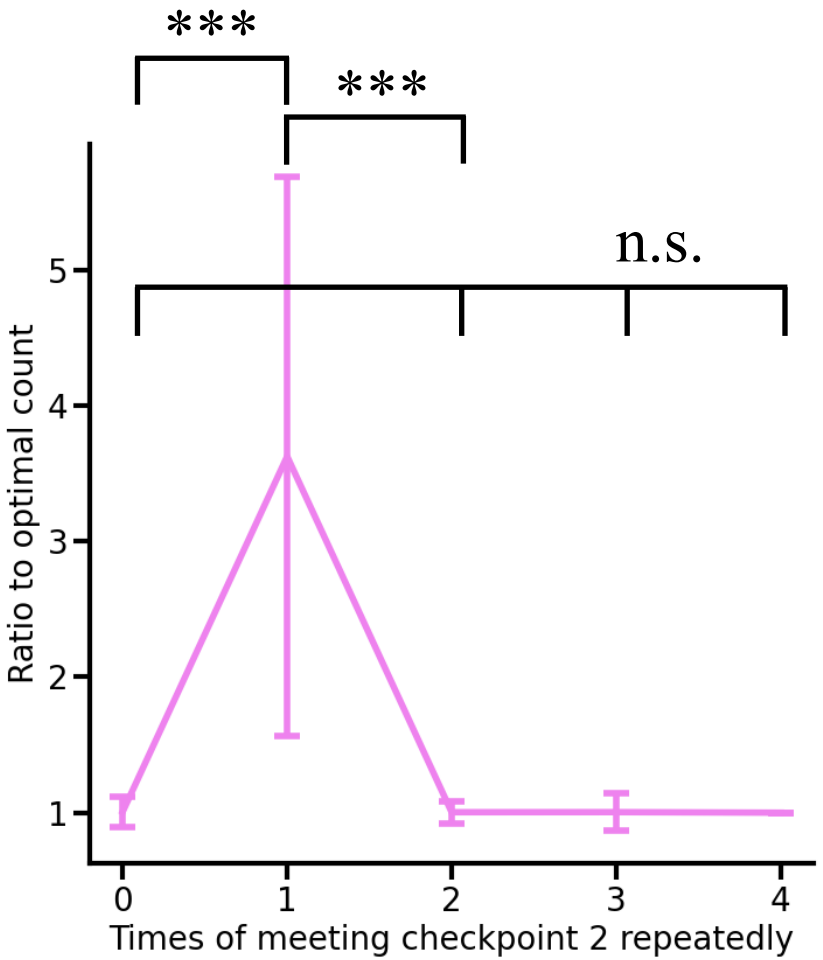
The mouse started from S, obtained a small reward at C1, and then reached G getting a big reward. We investigate changes in behavior and replayed trajectories when the reward switched from C1 to C2.







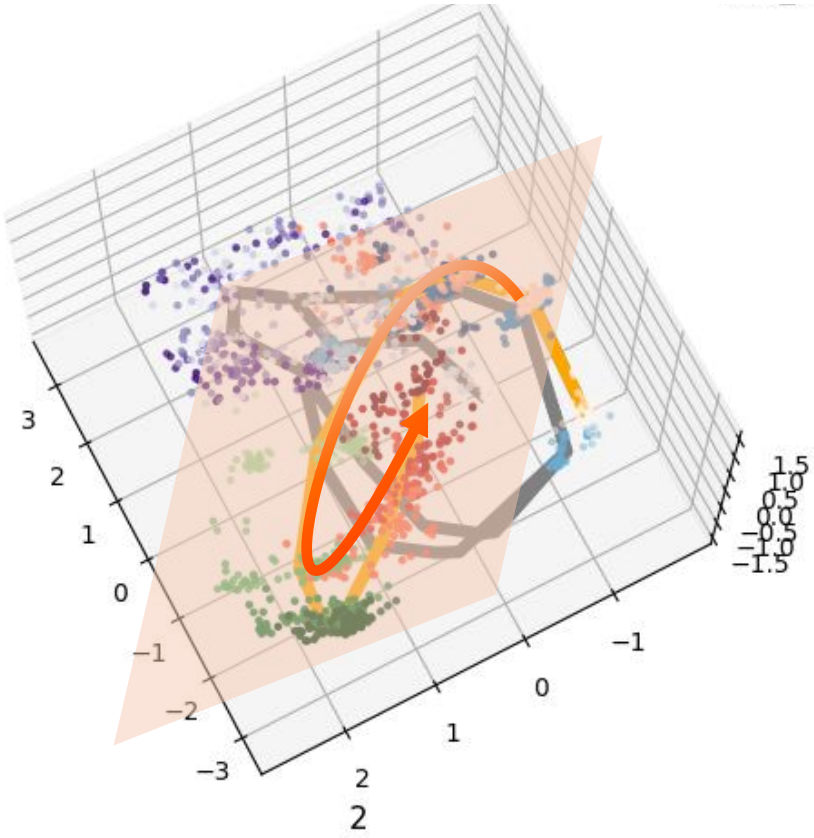
Proportion of replay trajectories representing different segments of the room.
Left: our distribution of segment replay as learning progresses.
Right: experimental results from Igata et al. 2021
High similarity in two aspects. First, the amount of replay of S-C2 increases.
Second, the amount of C2-G first increases and then drops down.



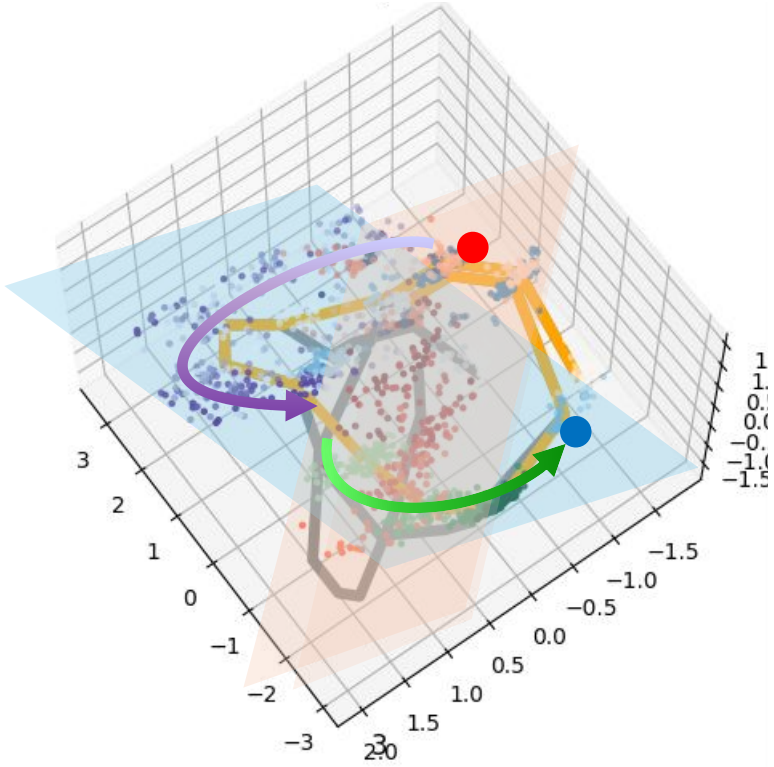
Performance of the AI agent when it comes across the new checkpoint, represented by the ratio of the number of agents' average steps to the optimal trajectory (in this task, the optimal number of steps is 3).

Manifolds: Principal components reduced from hidden states in PFC (3-dimension).

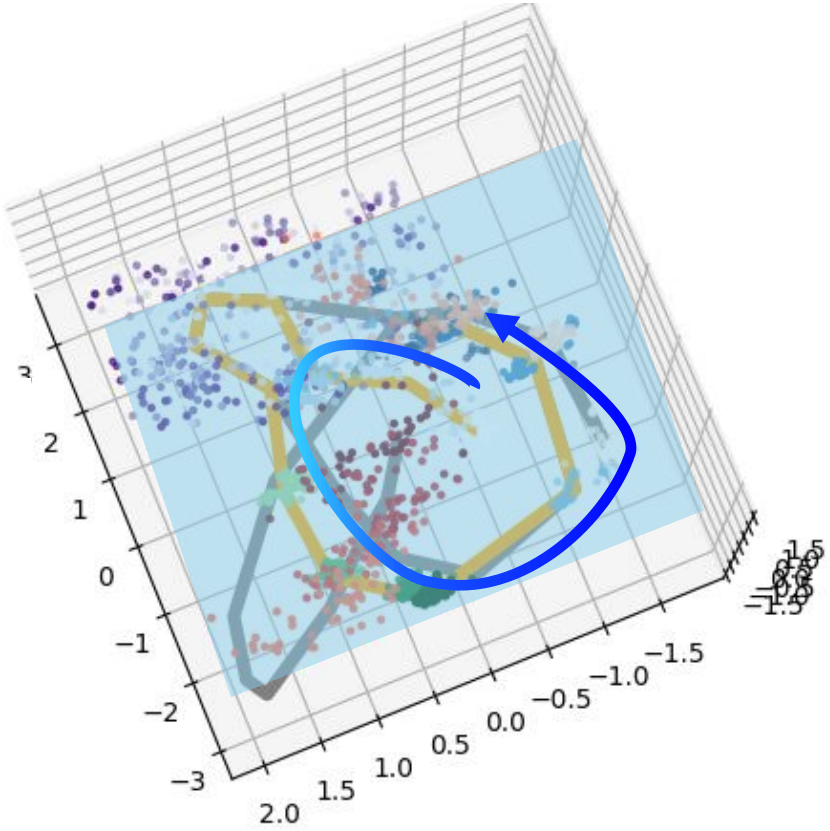
Reward at C1

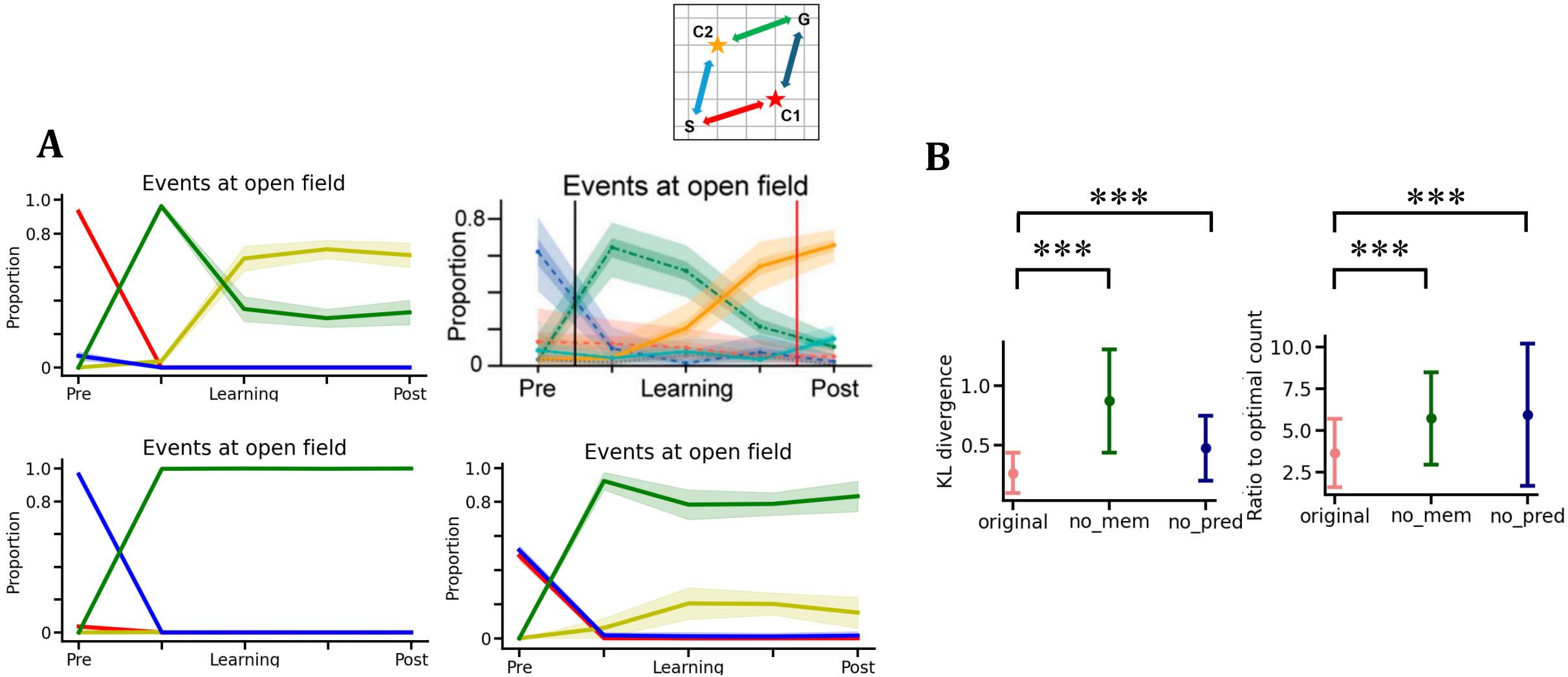


Transition



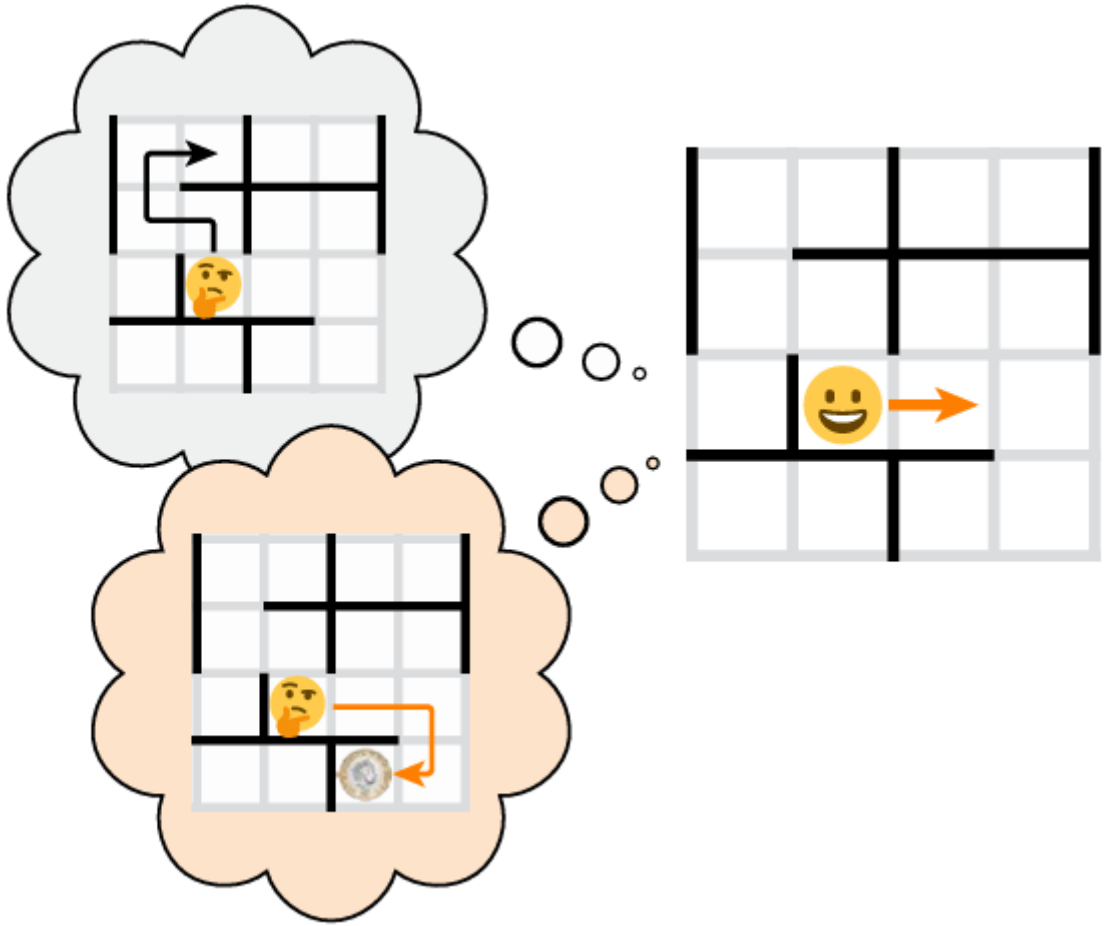
Reward at C2



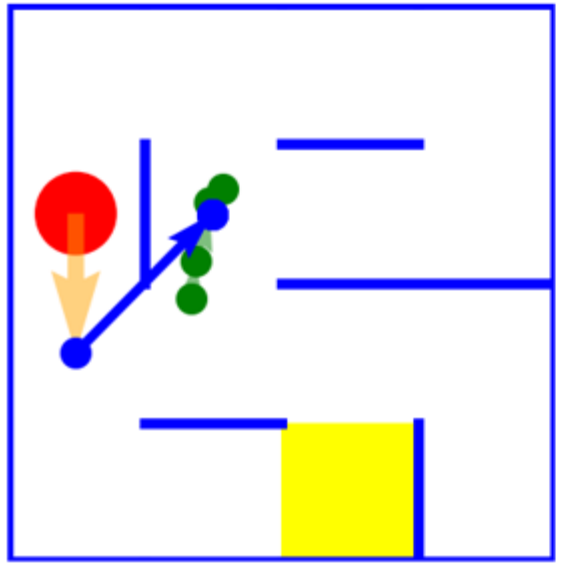


Ablation Studies: After removal of either the memory objective function or path-integrating objective function, the replay curve is less similar with the original experimental curve (A: bottom left, remove memory, bottom right, remove path integration; B: distribution similarities).
The main difference is that the replay sequences representing shortcut to new checkpoint doesn't increase, which indicates the failure of hippocampus activity to capture the new checkpoint.

Experiment 2



Replay is before the action



The current
HPC activation

Agent

Reward

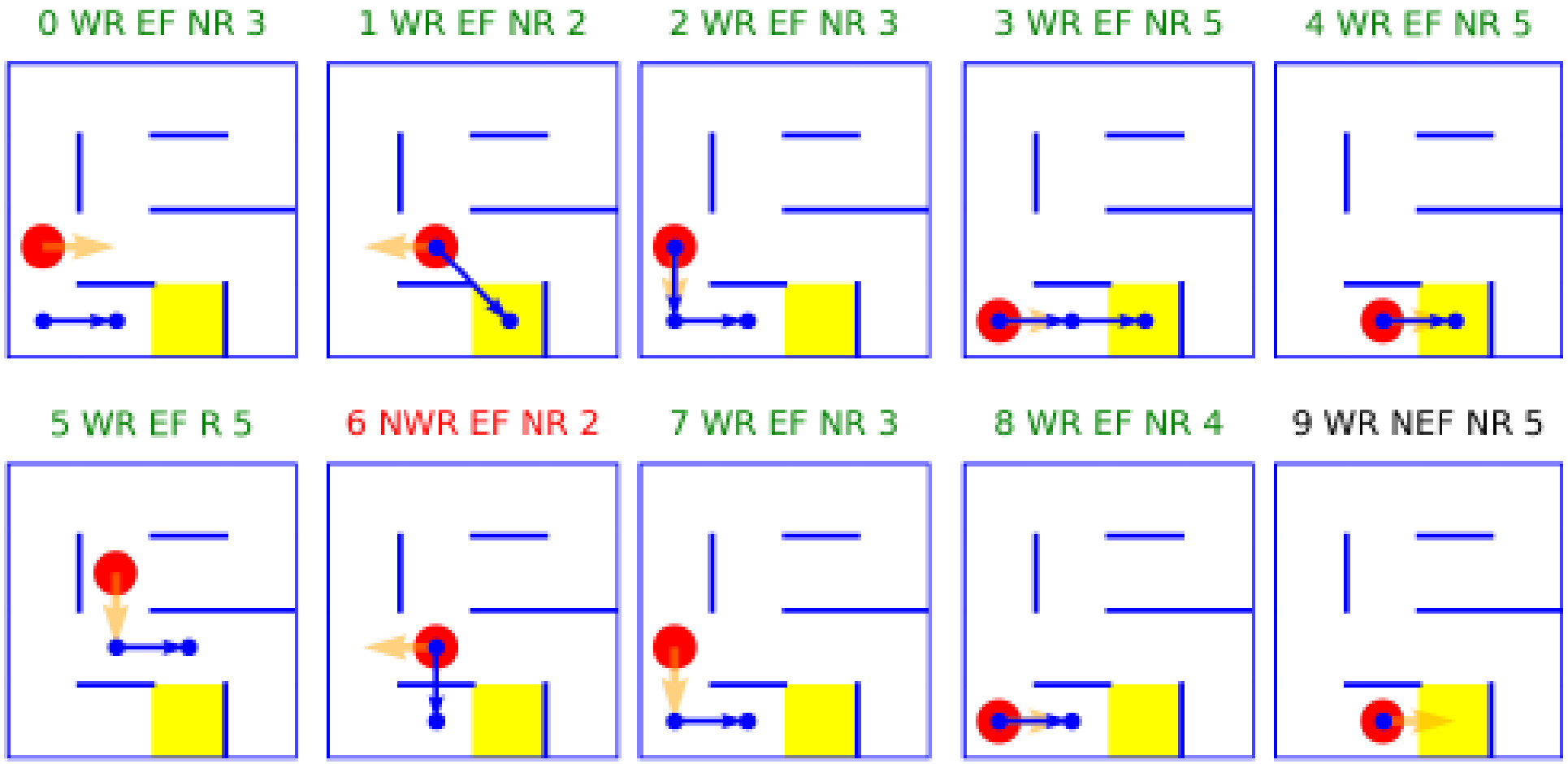
Next real action

Wall

Replay trajectory by argmax over the place cells

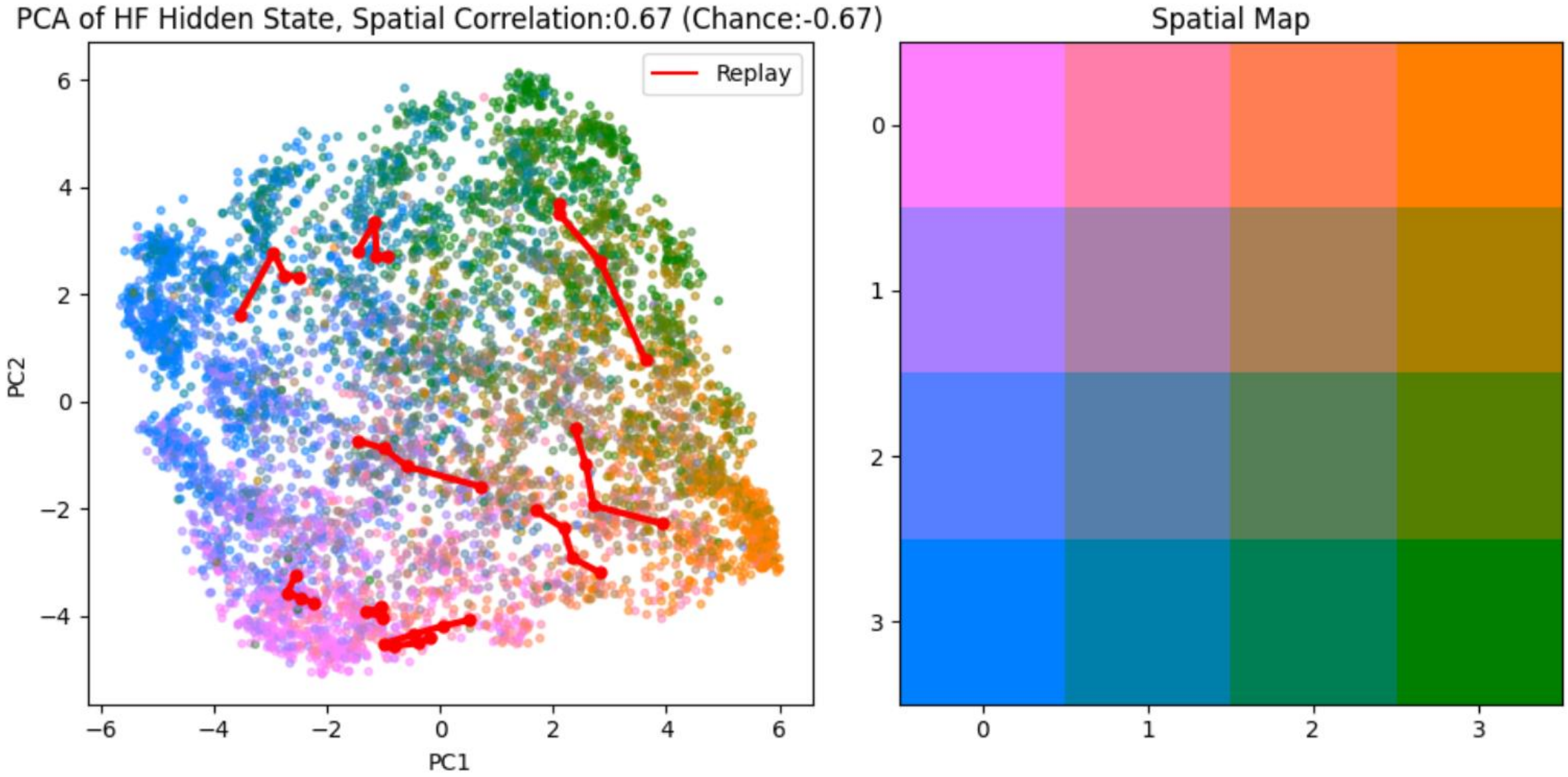
Replay trajectory by taking expectation over the place cells

Reset reward every 100 steps
Reset the environment every 10 times of resetting reward



63% of replays respect the wall.

Replay resides in the cognitive map

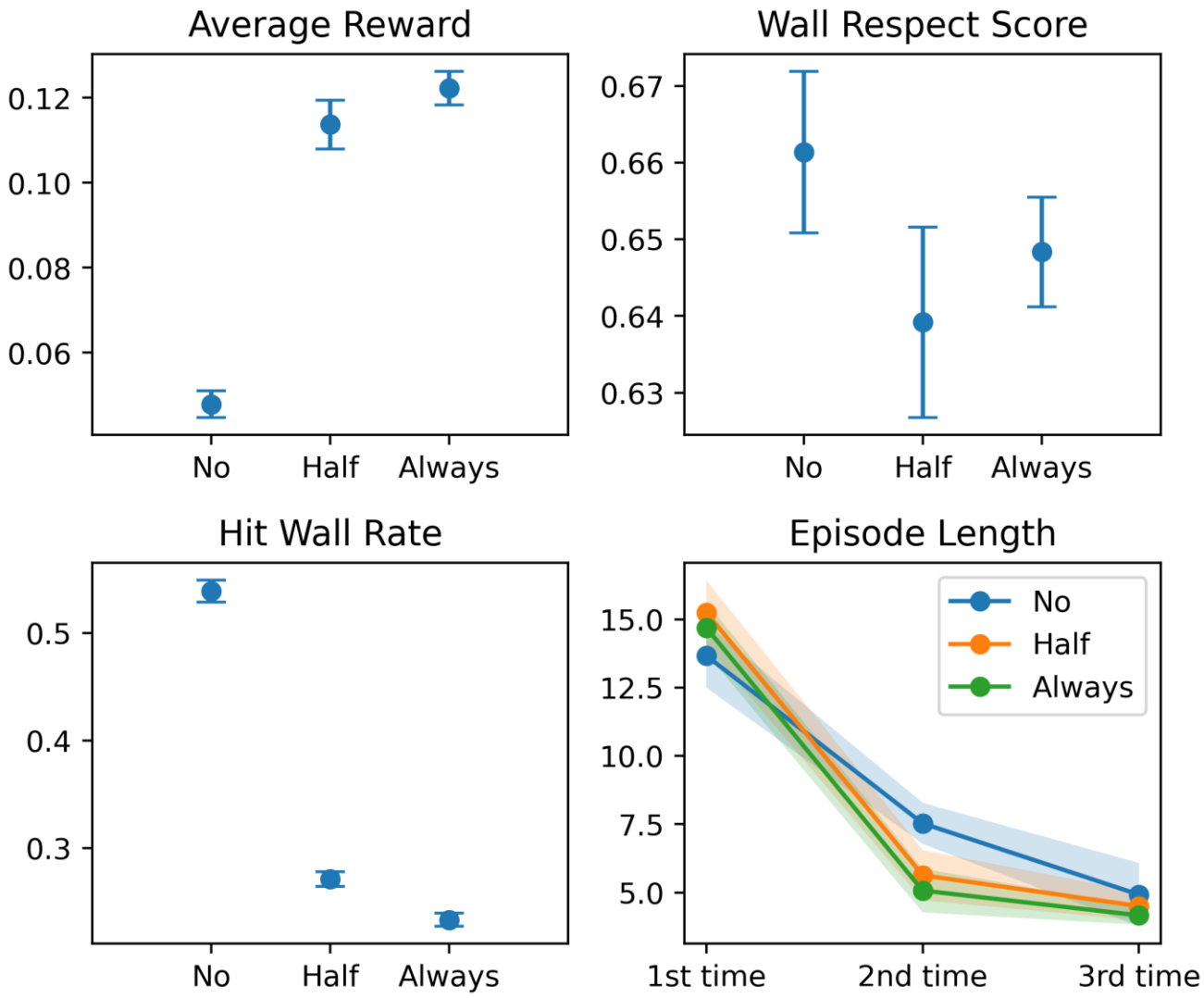


Replay improves the navigation efficiency

decision point

Train on the Half (50% replay communication) agent

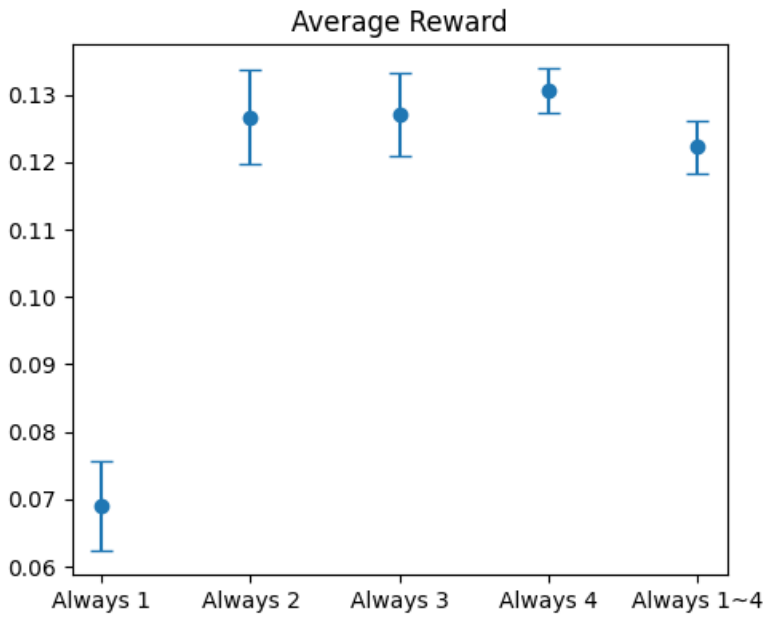
Test on No, Half and Always (100% replay communication) agents

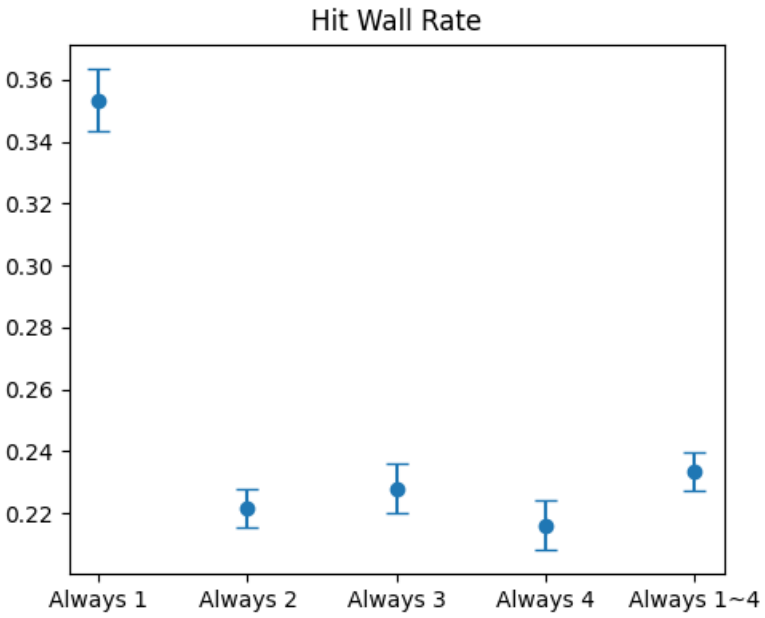


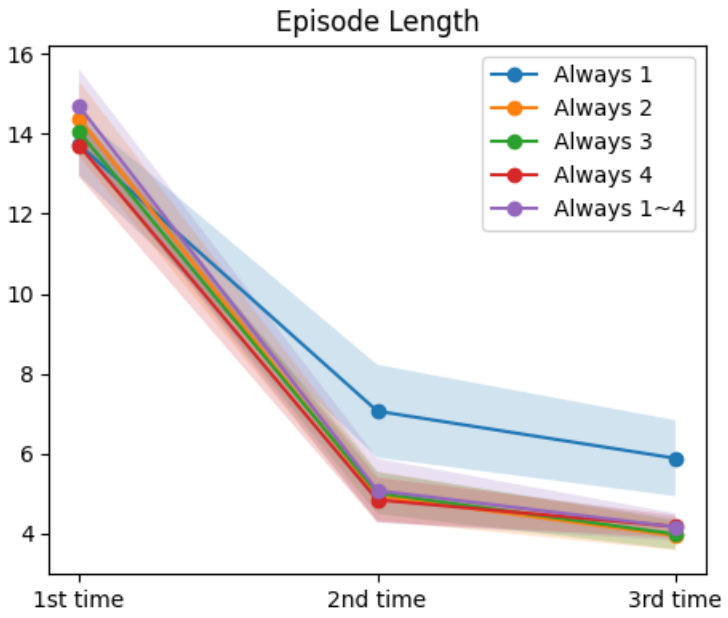
Wall Respect Score: 1- hit wall rate during replay

Longer replay is better? Saturated?

Need add a baseline of no replay

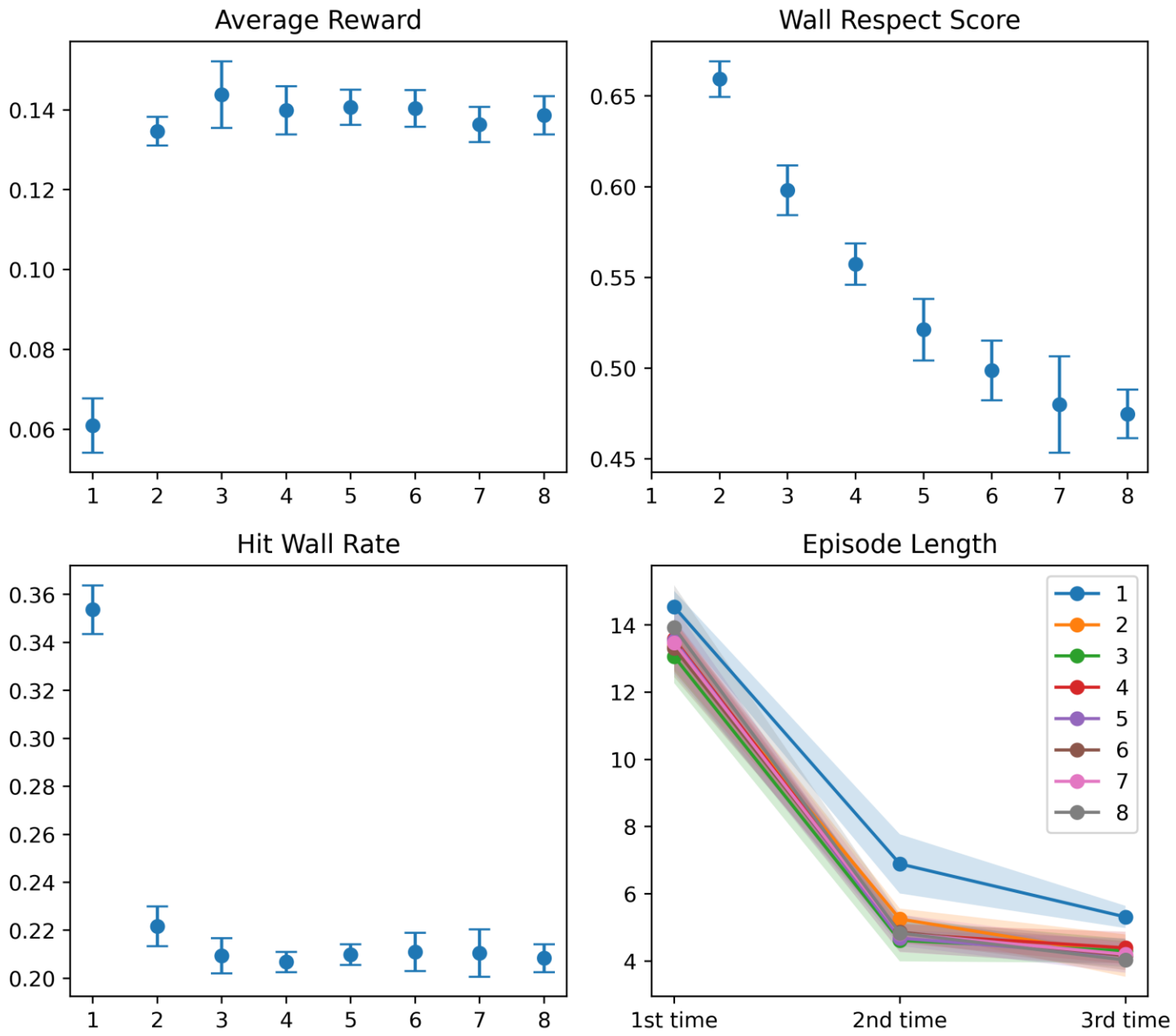






Train on the Half (50% replay) agent

Train on replay steps = 1~8



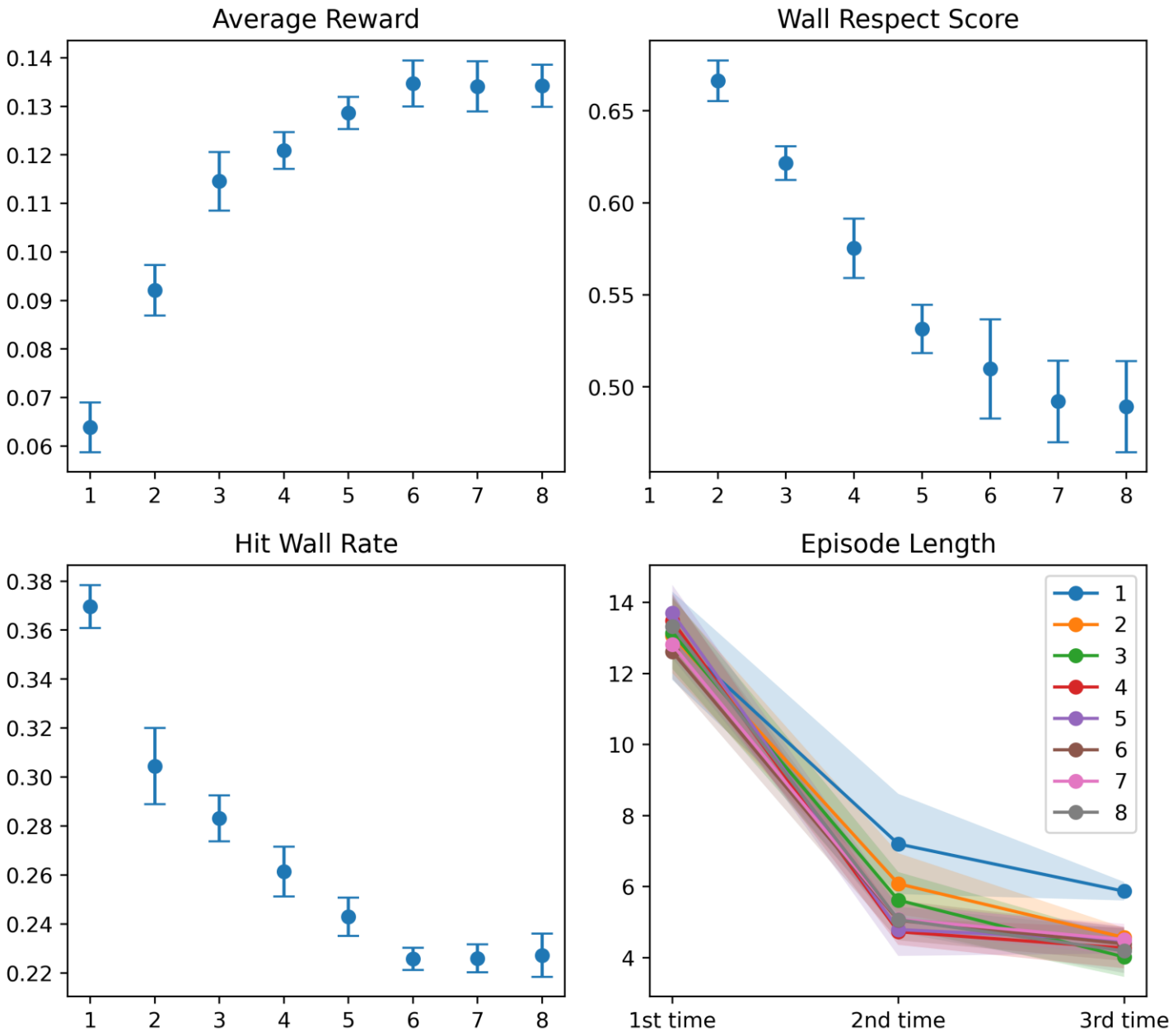
From replay steps=2, longer replay may not be better.

Maybe it's because we are training it on random steps of replay and the network learns to finish the computation in 2 steps.

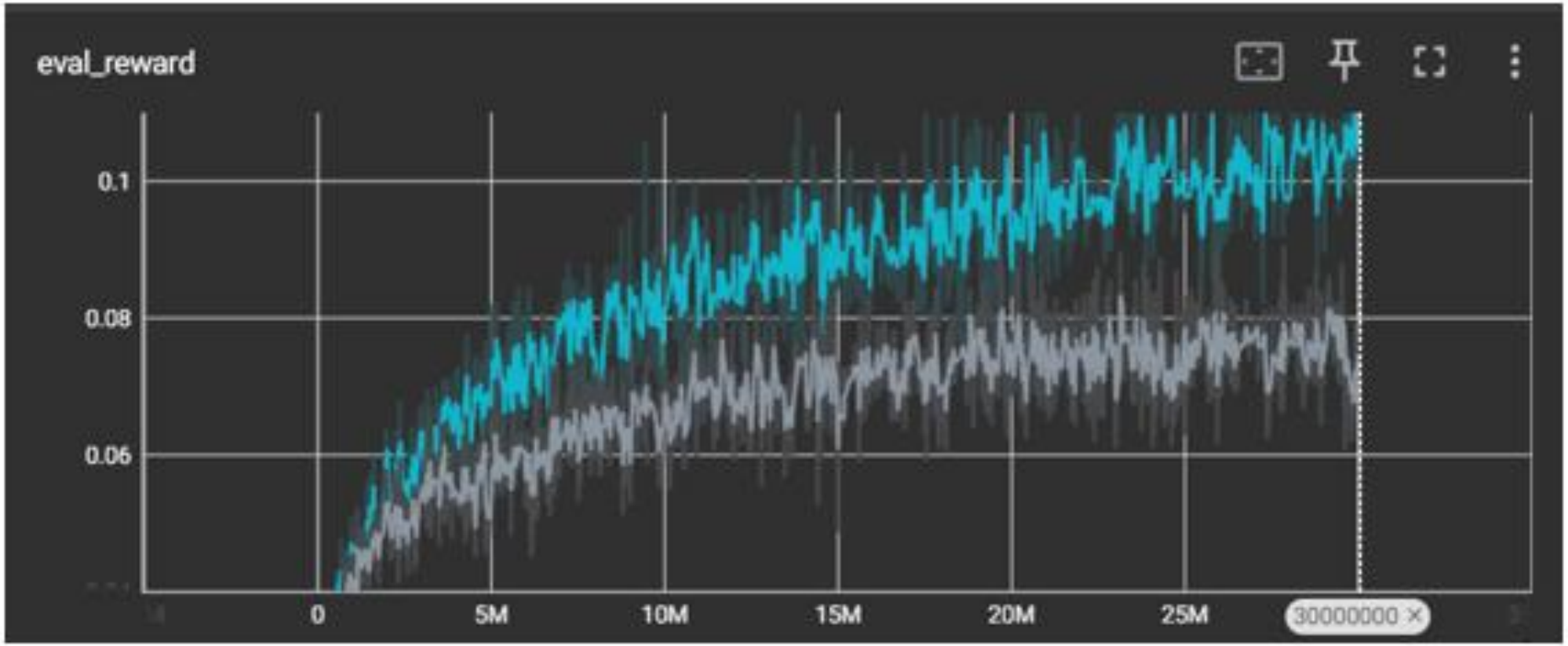
Train on replay steps = 4~8

add cost of time
choose how many steps

Alex Grave
ACT network adaptive time
"Adaptive computation time for recurrent neural networks." 2016.

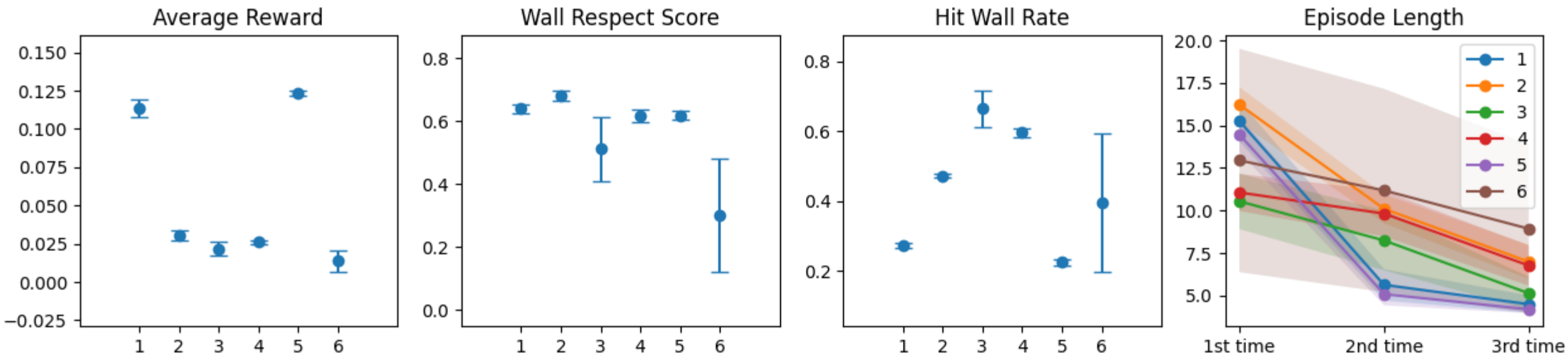


More frequent replay is better for training

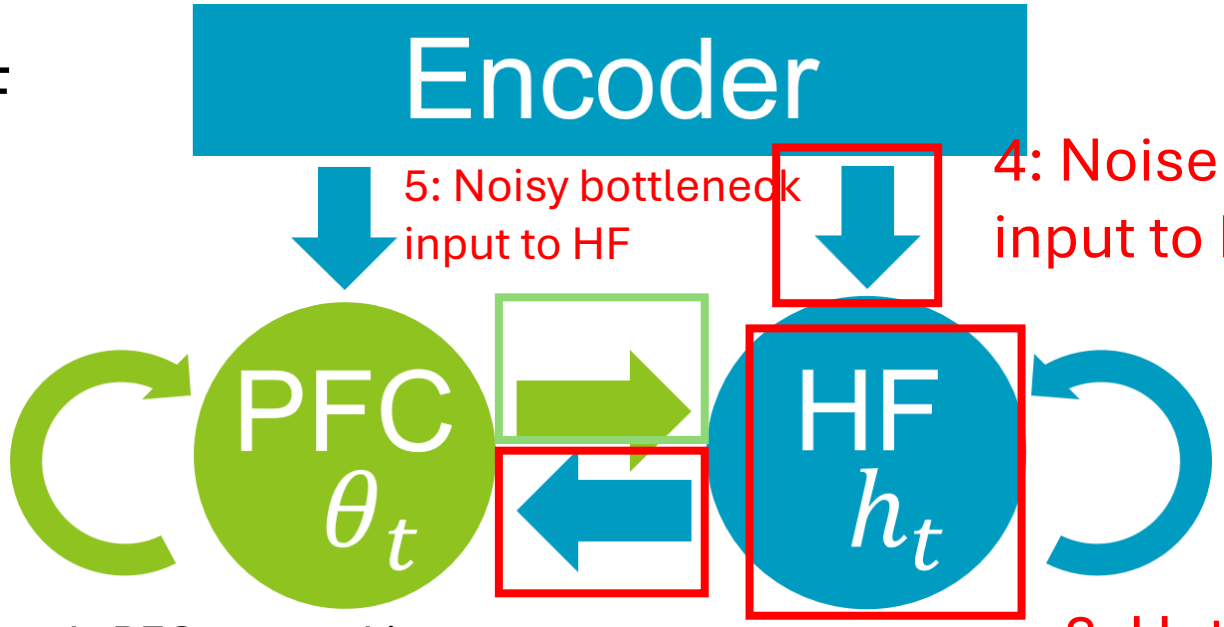


Always

Half



Ablations to HF impairs performance



5: Noisy bottleneck input to HF

4: Noise external input to HPC

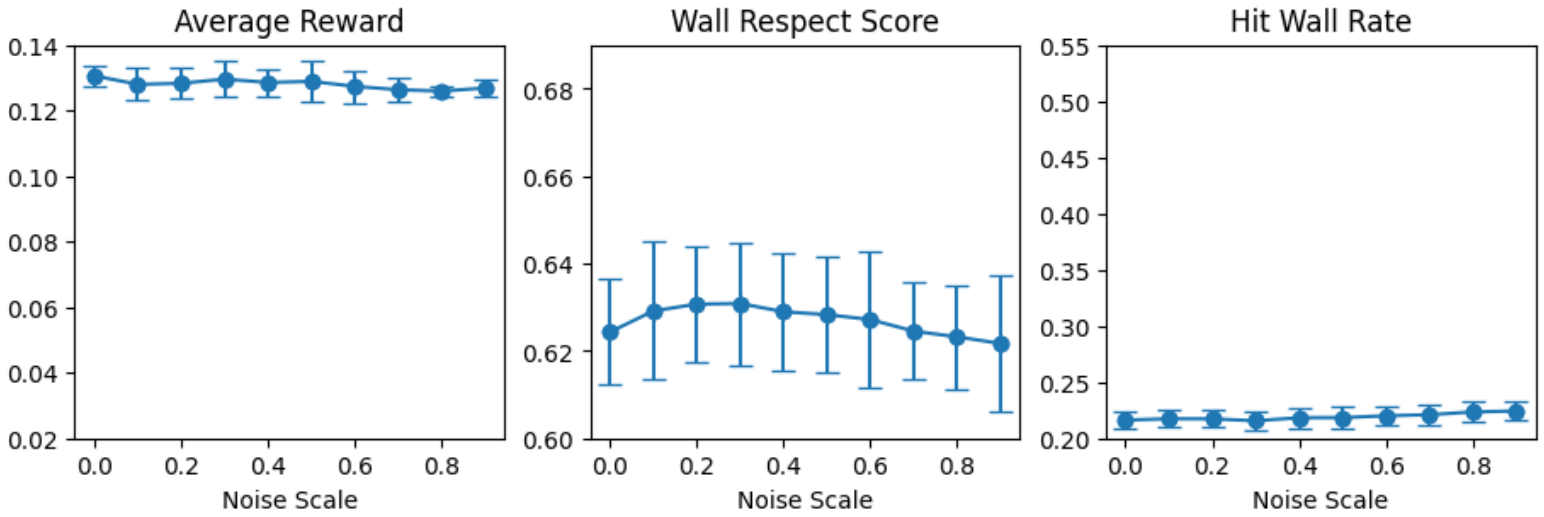
6: random agent before training

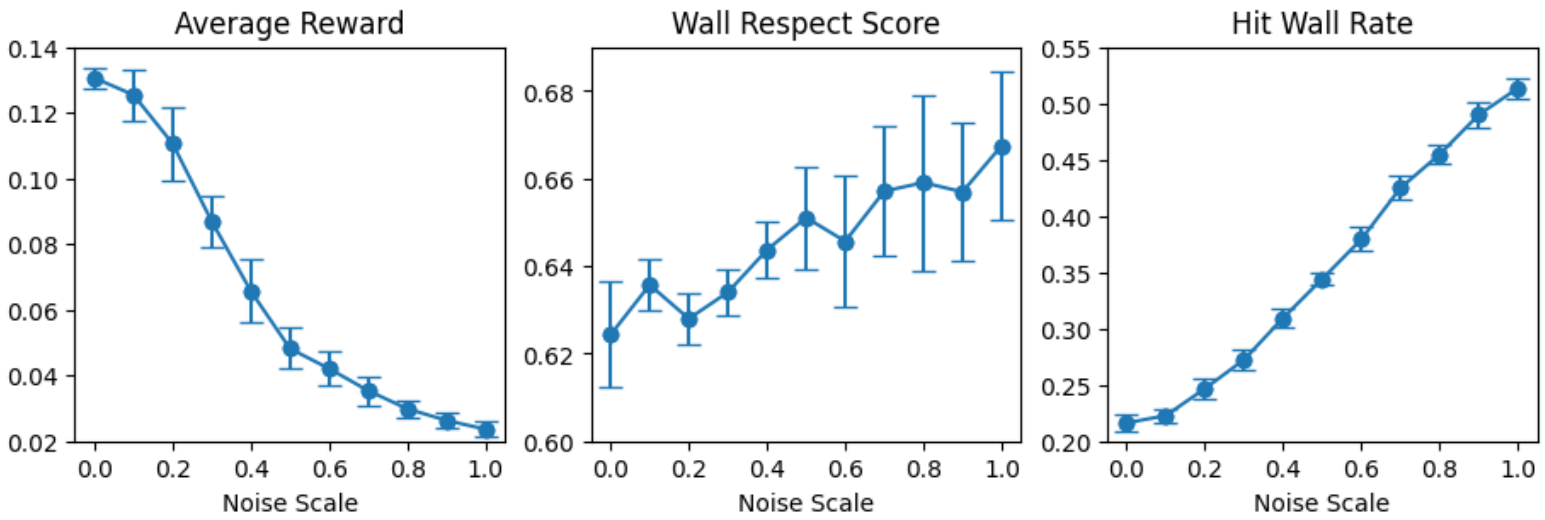
3: Untrained HF

2: Noisy bottleneck input to PFC

1: Original

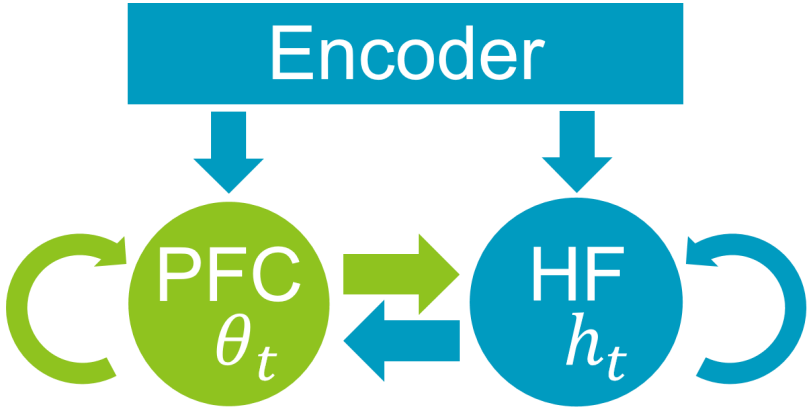
only PFC not working
aversion from walls?
representation analysis?





pfc_info input layer = 0?

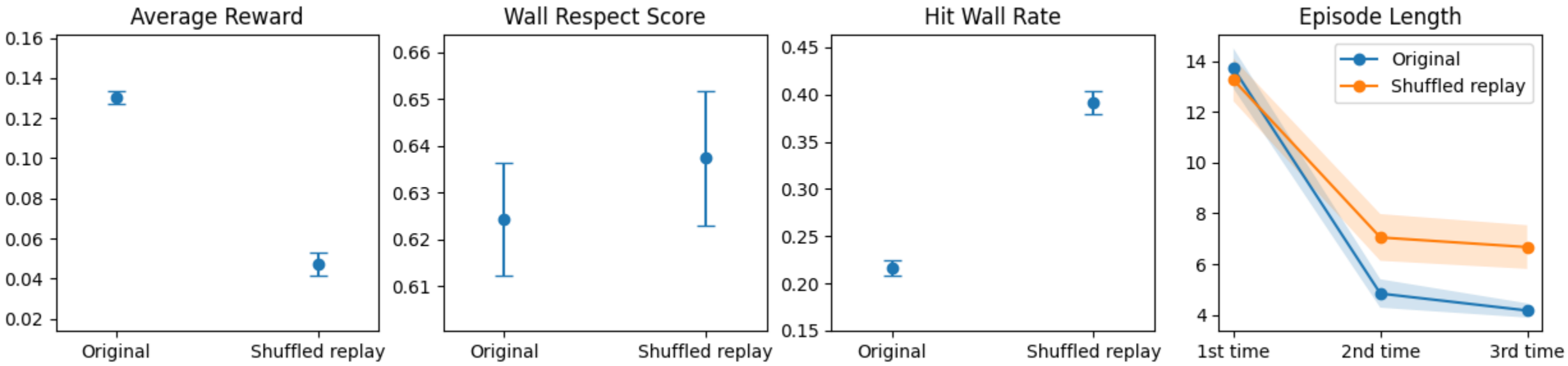
Noise to HF



Noise to PFC

Noise variance is matched at every replay step

Shuffled replay order within the same real-time step impairs performance



Always replay 4 steps

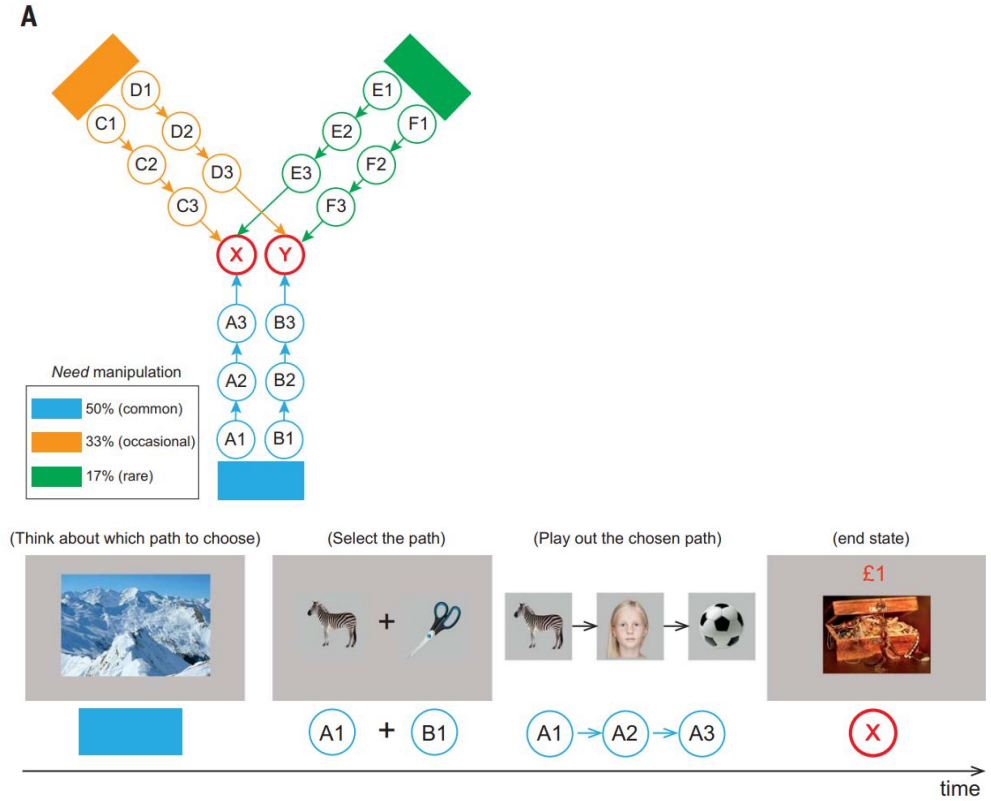
Different shuffle levels (while keep the replay order within the same real-time step) all impair performance

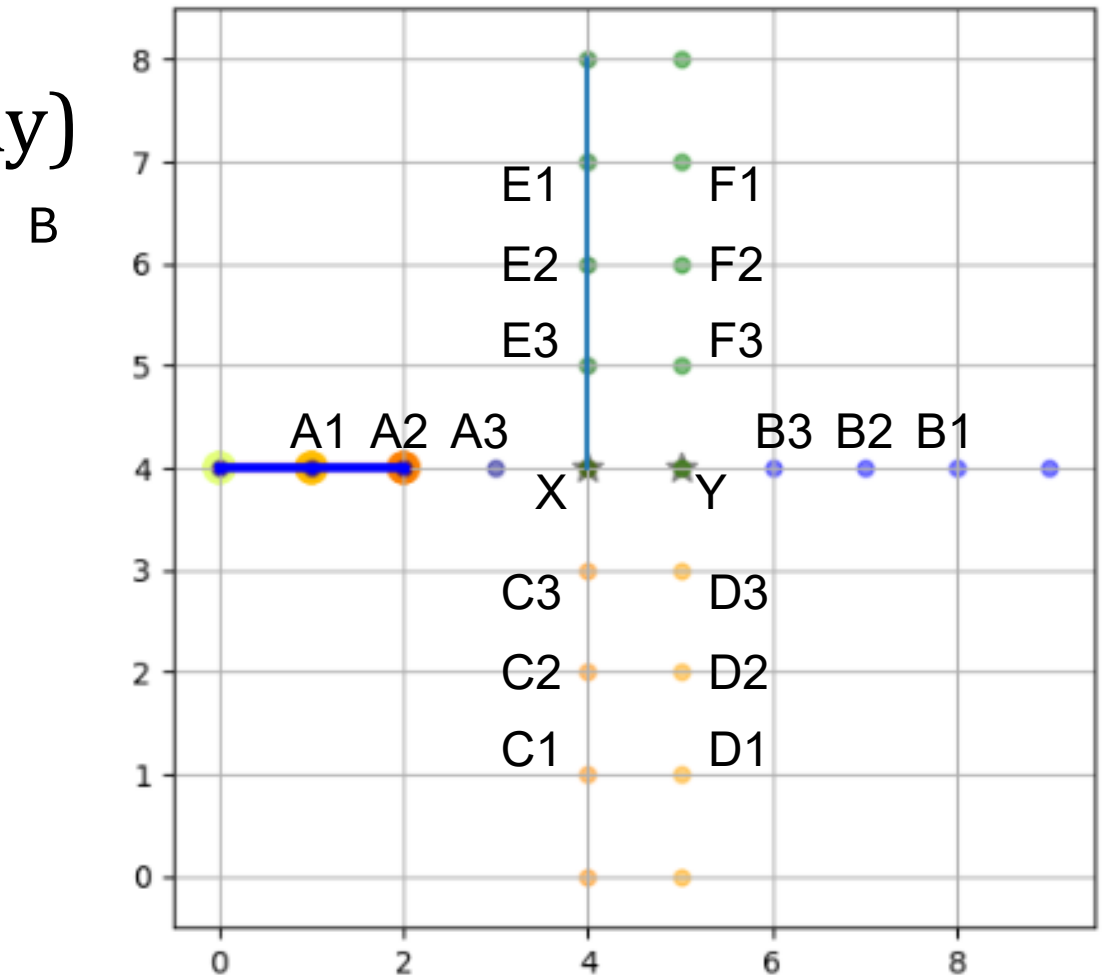
	test performance
seed0 replay info - seed0 test	0.12765625
seed0 replay info - seed1 test	0.0290625
seed0 replay info shuffled across the steps- seed0 test	0.029375
seed0 replay info shuffled within the same environment - seed0 test	0.0359375
seed0 replay info shuffled within the same goal setting and environment- seed0 test	0.04359375
seed0 replay info shuffled within the same location and environment	0.07515625

A seed is a randomly generated test situation

More Experiments

Multi-arm decision-making (Humans, backward/non-local replay)





A: Setting of adaptive decision-making task. Each dot represents a path state in (Yunzhe Liu et al. 2021). The states with the same color constitute an arm, and every arm has two segments determined by the choice of the agent. B: the light blue line indicates the real "trajectory"; the dark blue line indicates the replay sequences. Here, the agent is carrying out non-local replay.

Limitations & Next Work

•**Agent-decided rest/replay (ACT):** Let PFC (or HPC) decide *when* to rest and *how much* to replay—ratio and length adapt online, not fixed or randomly sampled.

•**Synaptic plasticity & transfer:** Add plasticity layers or dual-timescale parameters (fast weights/synaptic traces) so replay causes lasting weight updates, and test generalization across environments/multiple tasks.

•**Bi-directional HPC↔PFC pathway:** Explicitly model (CA3 →) CA1→ Subiculum→ NR→ mPFC and the reverse loop, testing whether Subiculum/NR merely relay or actively transform signals to coordinate decision making.

•**Modular hippocampal replay:** Decompose HF into submodules (e.g., CA3 generator, CA1 readout) and examine their distinct roles in initiating/formatting replay.

•**Computation during replay:** Analyze what computation replay performs (e.g., value update, trajectory simulation, representation reorganization) using decoding or representational analysis. What's the best way to detect that?